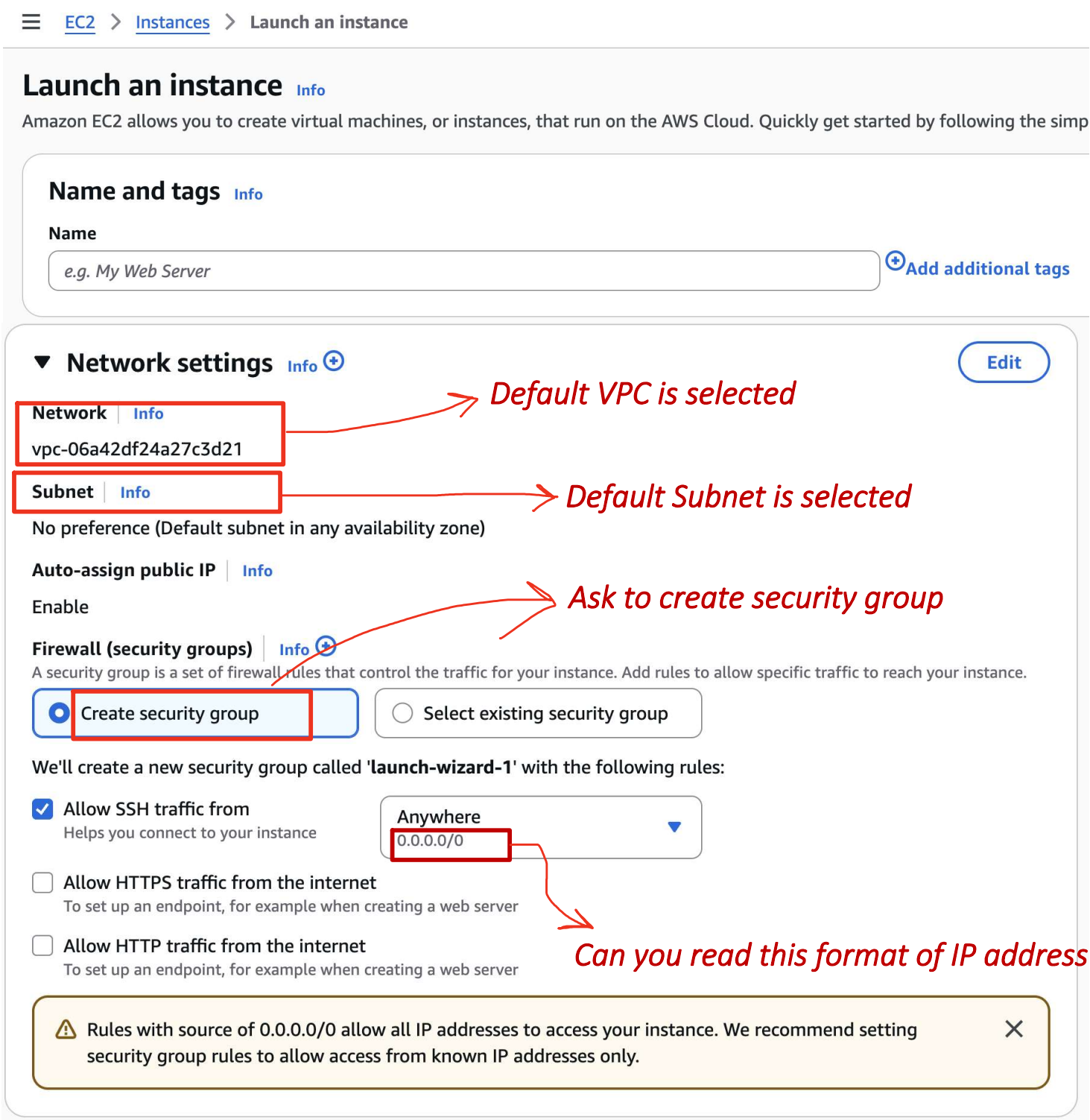


When we try to Launch **EC2** Instance, we will notice that most of the NETWORK details are set to default.



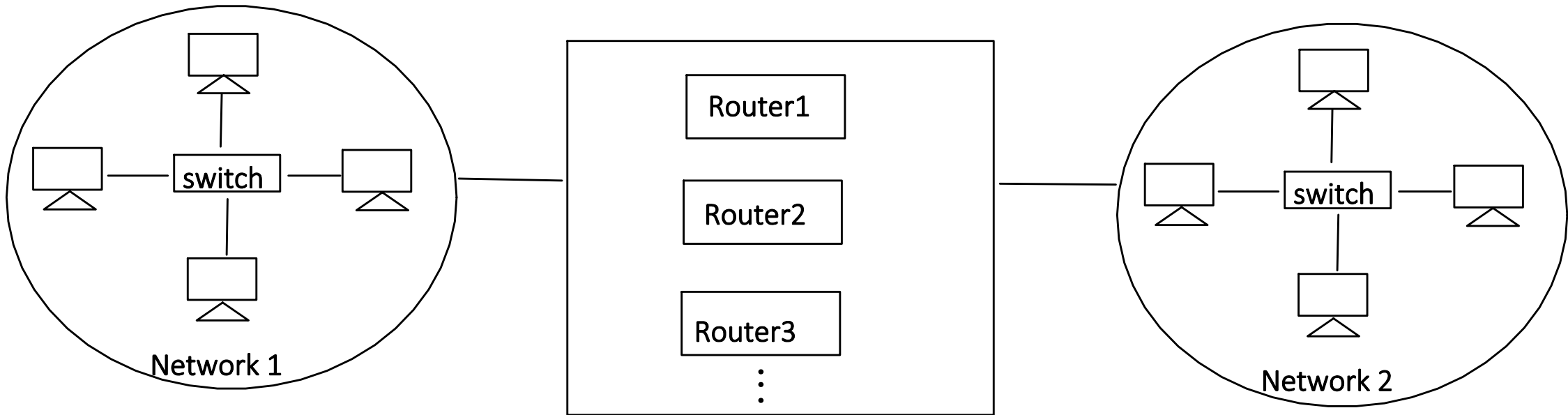
My recommendation would be not to proceed further without understanding NETWORKING concepts first.

Because, by using default Network configurations and by only setting "security group" we can launch **EC2** instances but it can introduce security risks and its not production-ready.

**Networking Fundamentals** before we start **AWS Network**

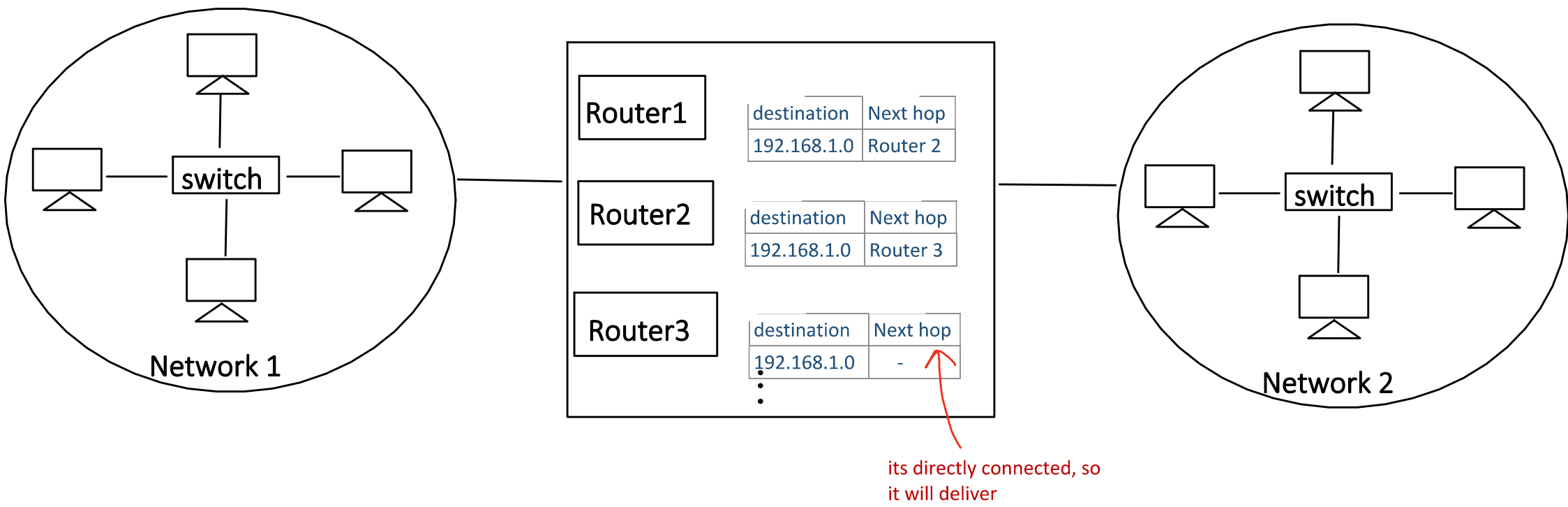
What is Network?

Group of hosts (computers) that can communicate with each other is called a network.  
2 different networks talk each other using routers.



Lets say, from Network 1 packet needs to reach to Network 2:

No single router knows how to reach from Network 1 to Network 2. Each Router hold the information about next hop and this detail is stored in "Routing Table"



IP Address:

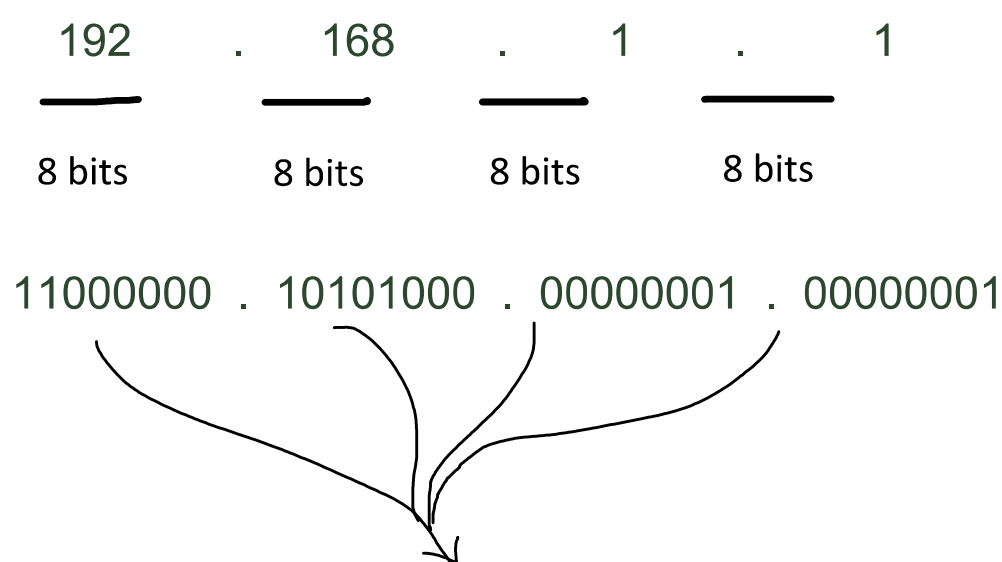
IP Address is the address of a device on a network, used for communication between devices.

We see IP address everyday, below is an example of IPv4

192.168.1.1



IPv4 is a 32 bit number.



Each can be a number between 0 - 255 ( $2^8 = 256$  possible values 0,1,2...255).

So, how many IPv4 address we can have:

$$\begin{aligned} 2^{32} \\ = 4,294,967,296 \\ = \text{approx. 4.3 billion addresses} \end{aligned}$$

There are tens of billions of devices which are connected to internet and need IP address.

So, yes we ran out of IPv4 address years ago and that's why IPv6 exists.

$$\begin{aligned} \text{which uses 128 bits} \\ = 2^{128} \\ = 3.4 * 10^{26} \text{ trillion addresses} \end{aligned}$$

So, no exhaustion problem at all. And good thing is AWS support IPv6 too.

But still, you will notice that most of the address (~99%) which we see day to day is IPv4 only.

So question is, if IPv4 addresses are exhausted years ago, then how internet still works with IPv4 addresses?

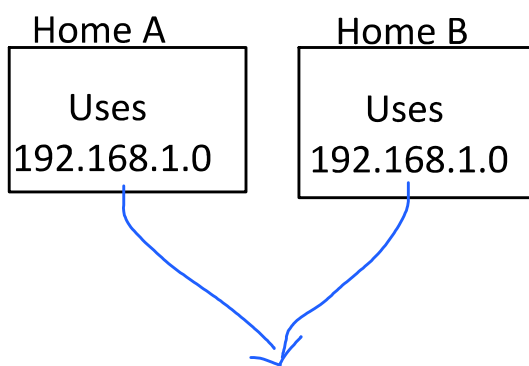
Answer: because of Private IPs

Below are the 3 private range:

| CIDR<br>(Classless<br>Inter-domain<br>Routing) | Description                                                                                                     | Lowest address                                         | Highest address                                                 | Total Available<br>IPs                      |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------|
| 10.0.0.0/8                                     | /8 means, left most 8 bits are fixed,<br>rest can vary.<br><br>00001010.00000000.00000000.00000000<br>(fixed)   | 10.0.0.0<br><br>00001010.00000000.00000000.00000000    | 10.255.255.255<br><br>00001010.11111111.11111111.11111111<br>1  | $2^{(32-8)} = 2^{24}$<br>~ 16.7 Million IPs |
| 172.16.0.0/12                                  | /12 means, left most 12 bits are fixed,<br>rest can vary.<br><br>10101100.00010000.00000000.00000000<br>(fixed) | 172.16.0.0<br><br>10101100.00010000.00000000.00000000  | 172.31.255.255<br><br>10101100.00011111.11111111.11111111<br>1  | $2^{(32-12)} = 2^{20}$<br>~ 1 Million IPs   |
| 192.168.0.0/16                                 | /16 means, left most 16 bits are fixed,<br>rest can vary.<br><br>11000000.10101000.00000000.00000000<br>(fixed) | 192.168.0.0<br><br>11000000.10101000.00000000.00000000 | 192.168.255.255<br><br>11000000.10101000.11111111.11111111<br>1 | $2^{(32-16)} = 2^{16}$<br>~ 65,536 IPs      |

What makes them Private?

1. No public router on the internet will forward packets to these addresses. The internet pretends these addresses does not exist.
2. Anyone can use them simultaneously without conflict.



Both hosts uses same private IP address without any conflict.

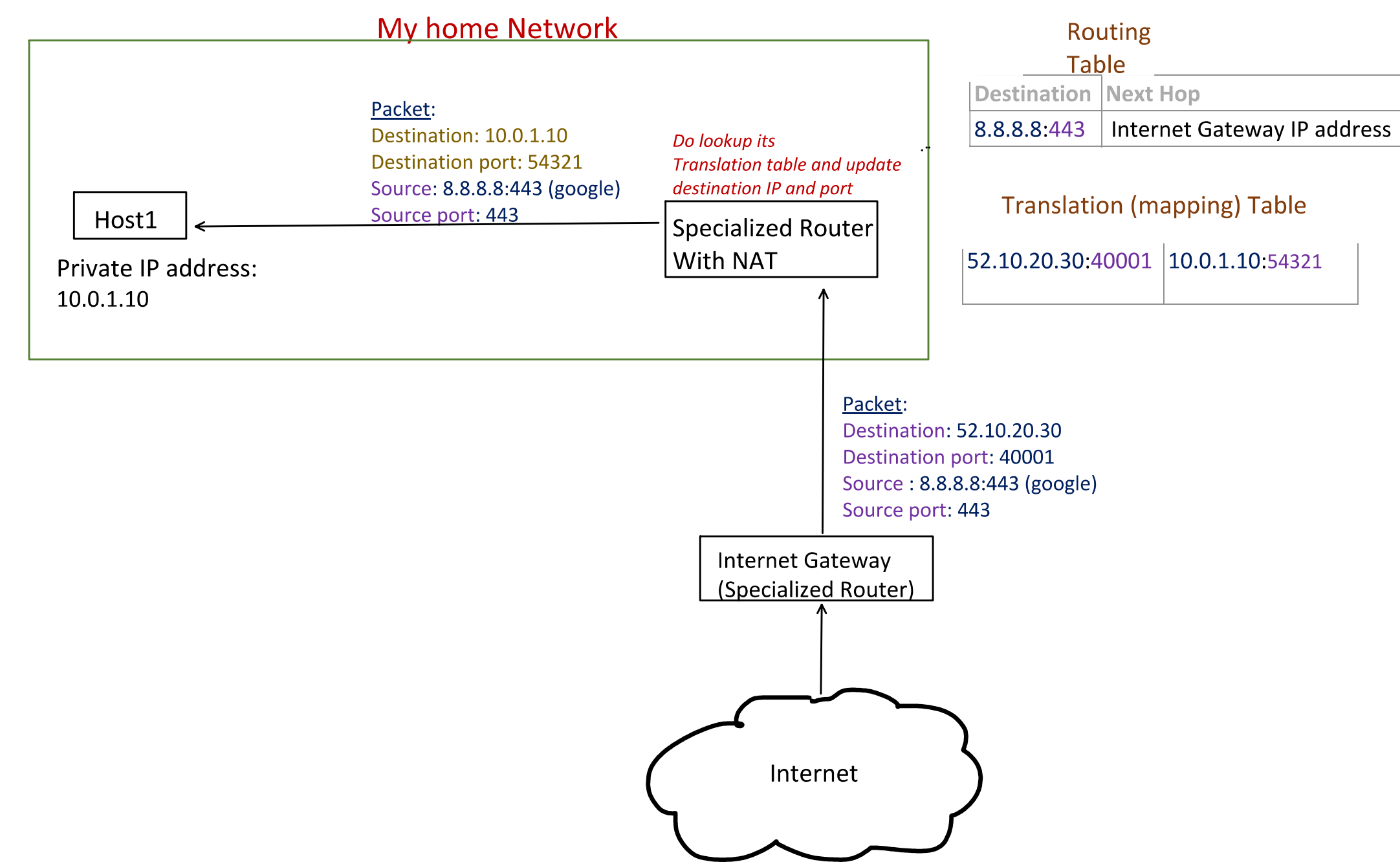
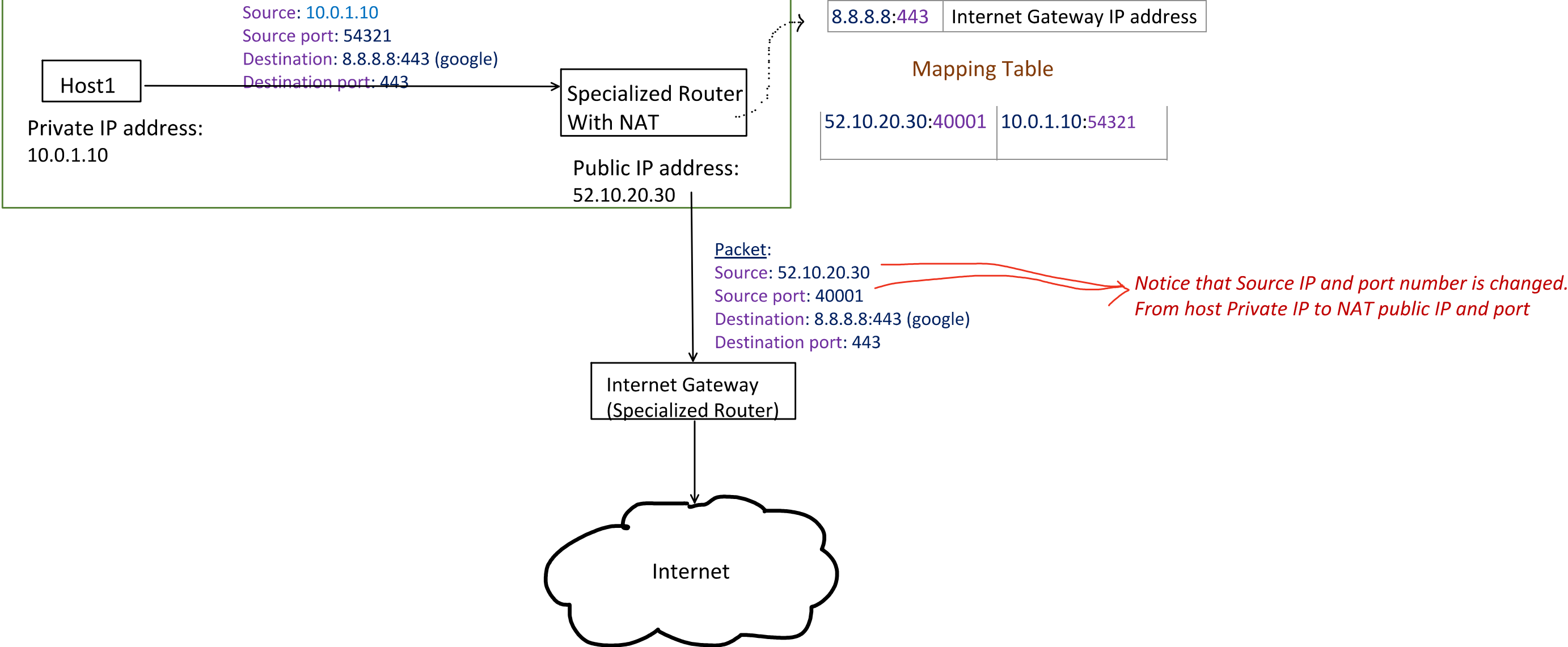
Now next obvious question comes to mind is: How private host reach the internet?

through, NAT (Network Address Translation)

Consider NAT is a software which runs on router that modifies the IP address while they are in transit.







At this point of time, many time confusion comes to connect the dots between:

- Network
- Private IPv4
- Switch
- Router
- Routing table
- NAT (specialized router)
- Internet gateway (specialized router)

Lets try to see whole flow connecting everything together:

#### 1. Allocate IP addresses for our Network

Requirement is max 500 devices (hosts) can be within this network.

CIDR I am choosing is: 192.168.0.0/16

11000000 . 10101000 . 00000000 . 00000000

(fixed) (available for hosts)

Total no of IP address available is:

$$= 2^{(32-16)} = 2^{16}$$

~ 65,536 IPs

But our requirement is only 500 devices, so we can further reduce the count:

192.168.0.0/23

11000000 . 10101000 . 00000000 0 . 00000000

(fixed) (available for hosts)

Total no of IP address available is:

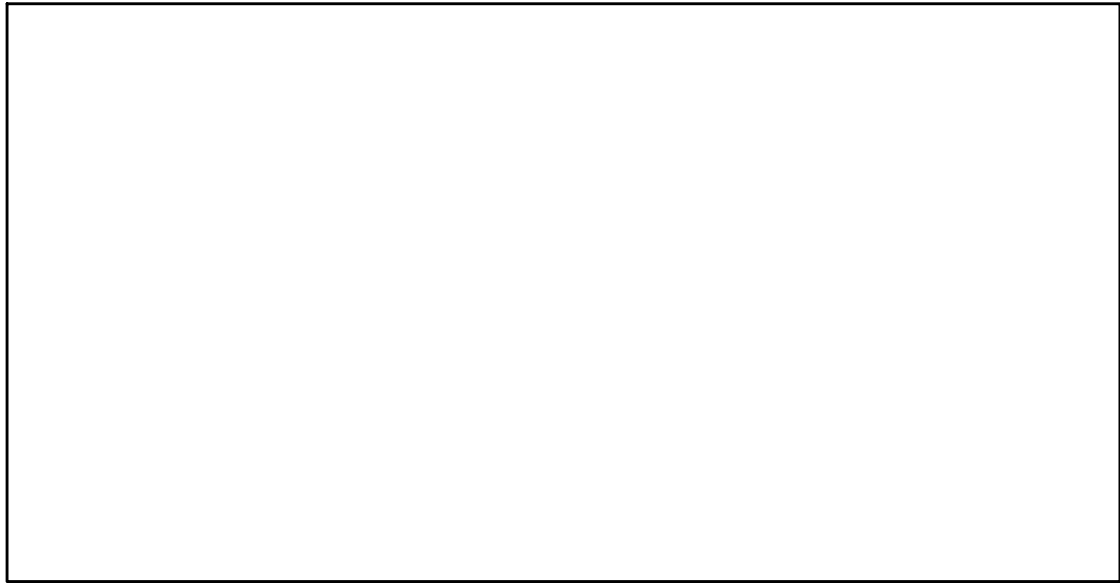
$$= 2^{(32-23)} = 2^9$$

~ 512 IPs

1. The fixed part 11000000 . 10101000 . 00000000.00000000 identifies our Network. Means all devices within this network starts with this only.
2. The second part 11000000 . 10101000 . 000000000.00000000 identifies the particular host within this network.



Network:  
CIDR: 192.168.0.0/23 (192.168.0.0 - 192.168.1.255)



1st address: 192.168.0.0 is reserved - known as NETWORK address

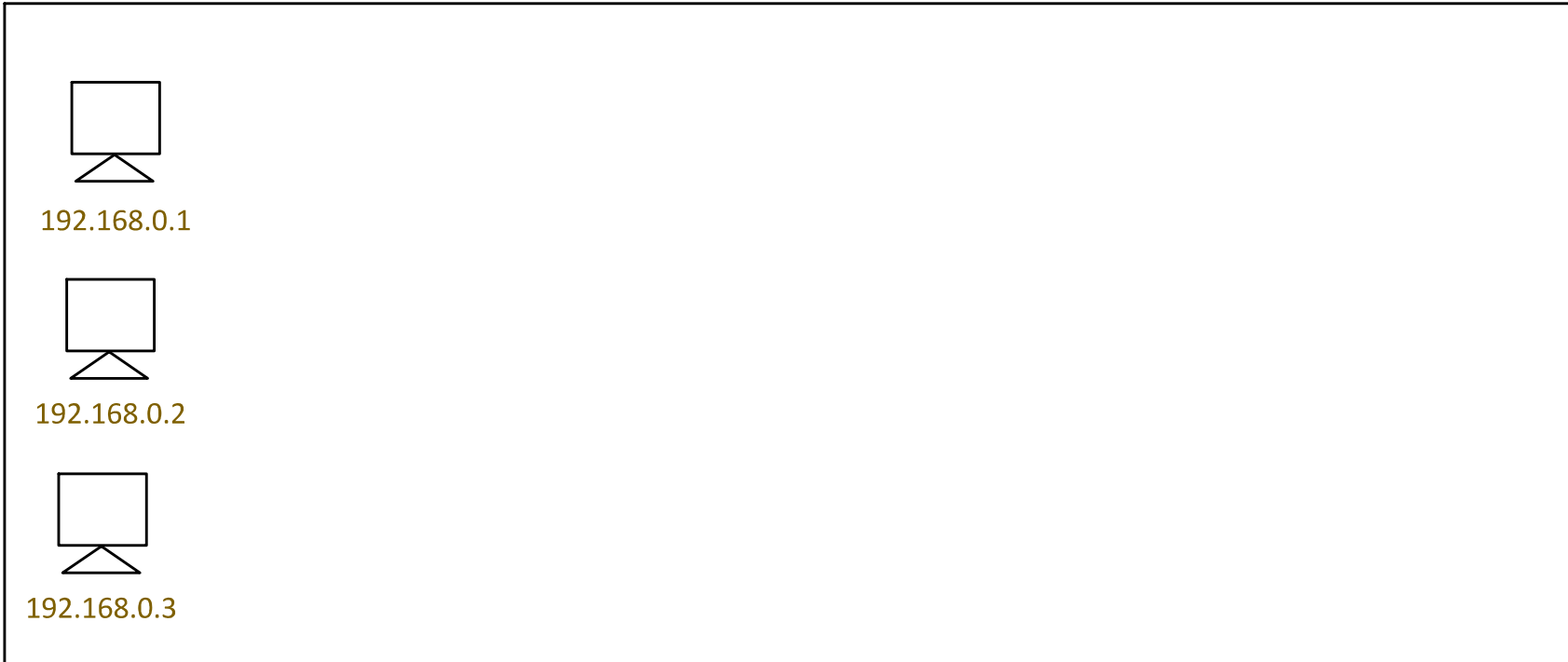
Last address: 192.168.1.255 is reserved - known as BROADCAST address

Rest all can be used by hosts.

2. Hosts within the Network

Each host which need IP address is allocated from the CIDR range.

Network:  
CIDR: 192.168.0.0/23

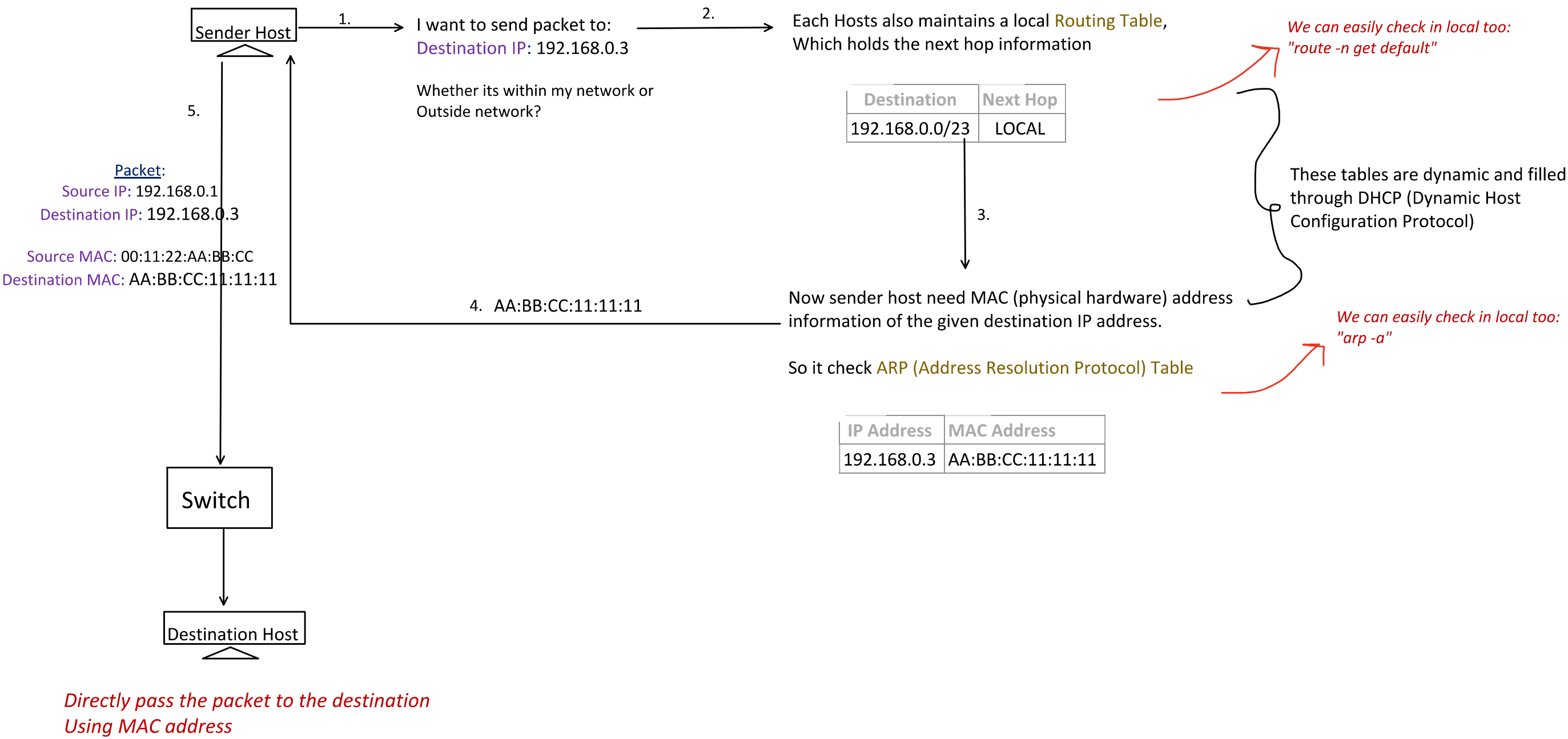


3. Hosts within Network are connected via Switch

Network:  
CIDR: 192.168.0.0/23

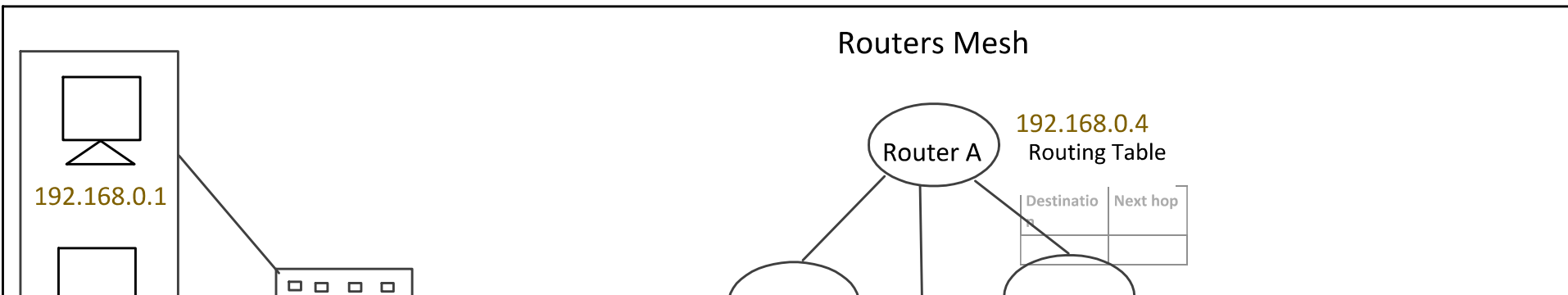


Remember "Switch" always work with MAC address not IP Address.

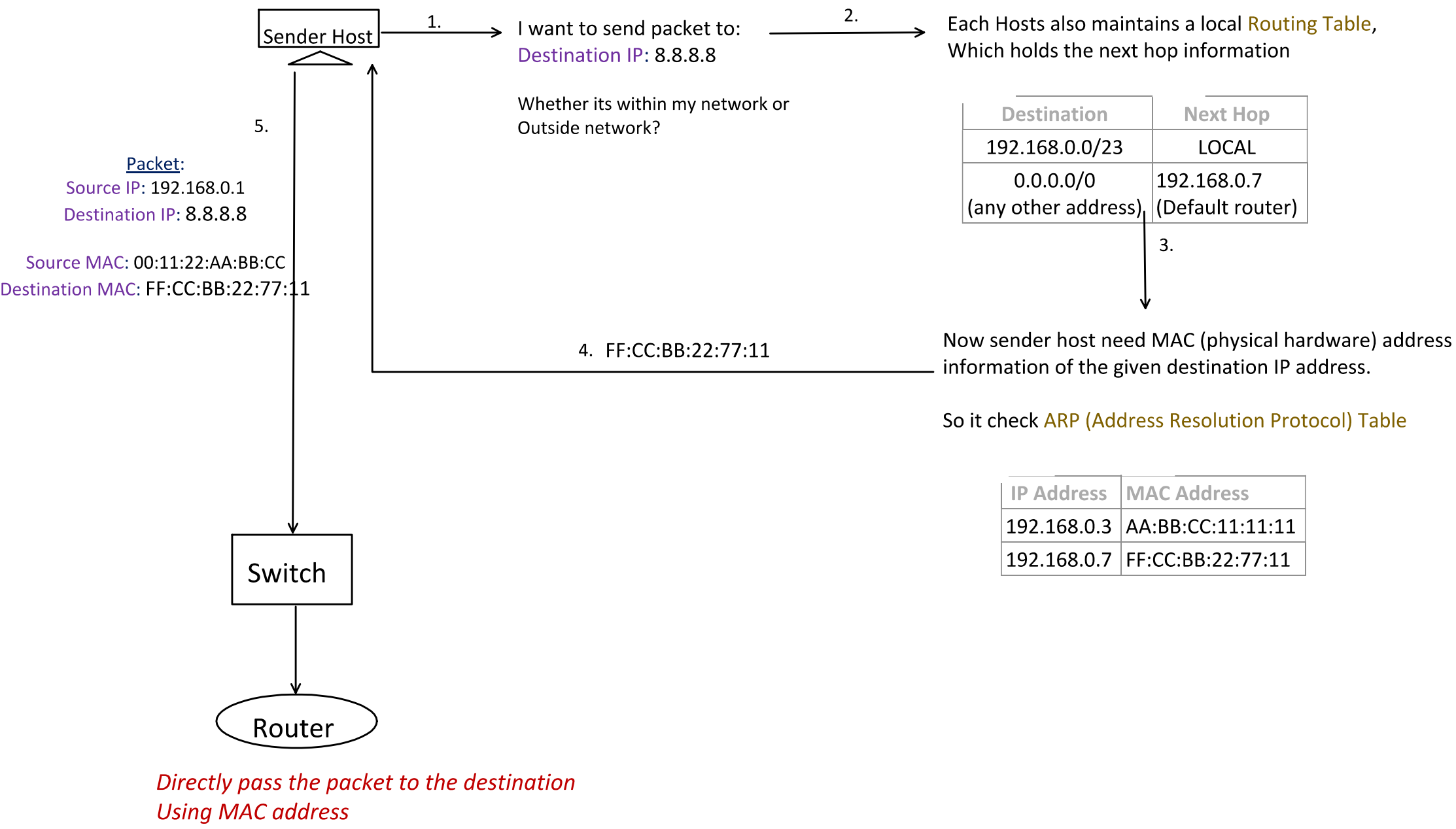
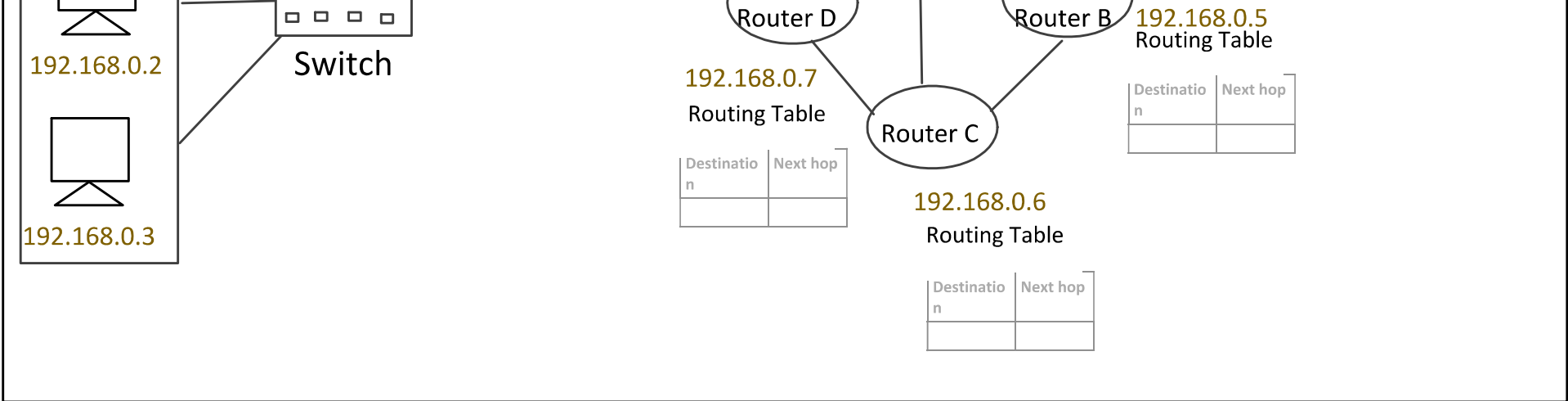


4. Routers

Network:  
CIDR: 192.168.0.0/23





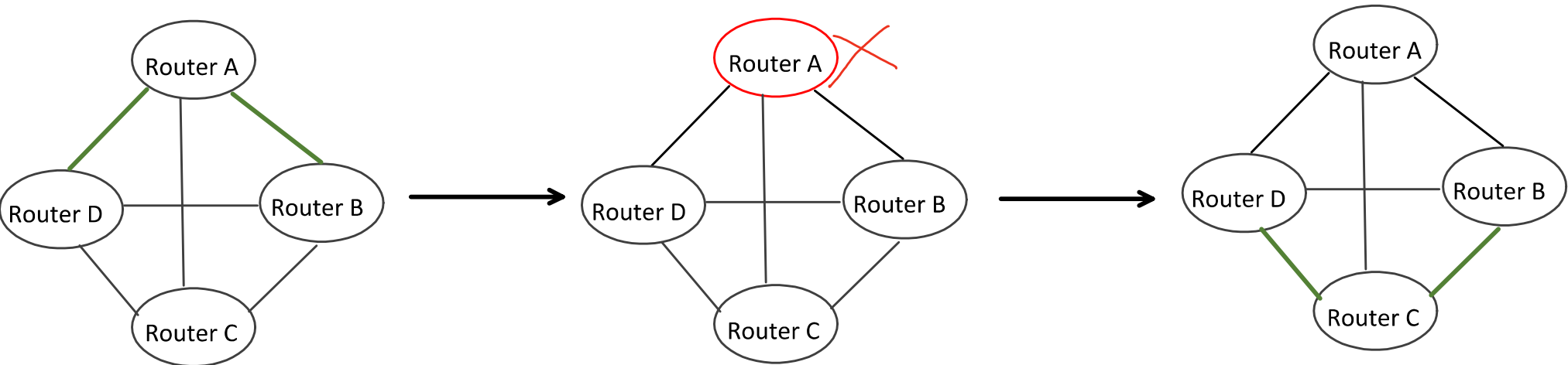


One doubt might come is, why we need Router Mesh:

Couple of reasons:

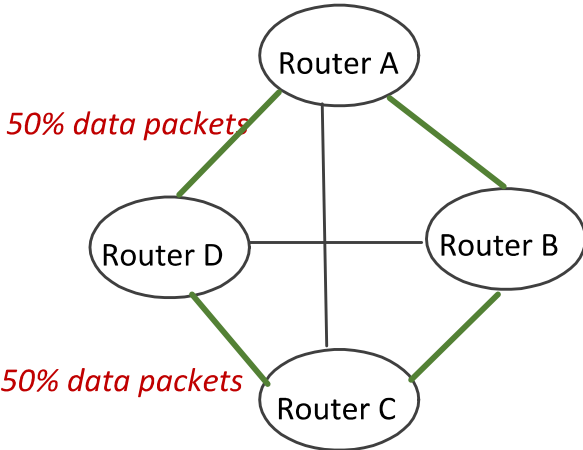
1. Fault Tolerance:

Because of Mesh, even if 1 router failed, packet will still continuously flow with other path.



2. Load Balancing:

If Router-D need to send huge influx of data packet to Router-B, it doesn't have to choke single path. It can send say 50% via Router-A and 50% via Router-B



3. Long distance handling:

Say destination is present in different network which is very far.

And everything device has a limit, so possible that 1 router is not sufficient and we need multiple routers to forward the packet before it reach the destination.

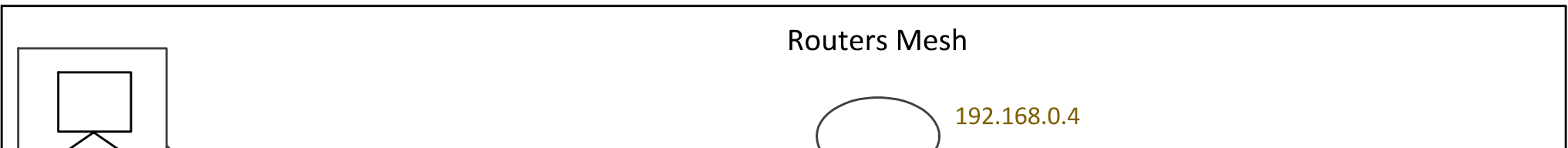
5. NAT

- Removes sender private IP (192.168.0.1) and replaces it with its own **Public IP**.
- Sends it out through the Internet Gateway to the web.
- When the response comes back, the NAT remembers who originally asked for it and forwards it back down the chain.

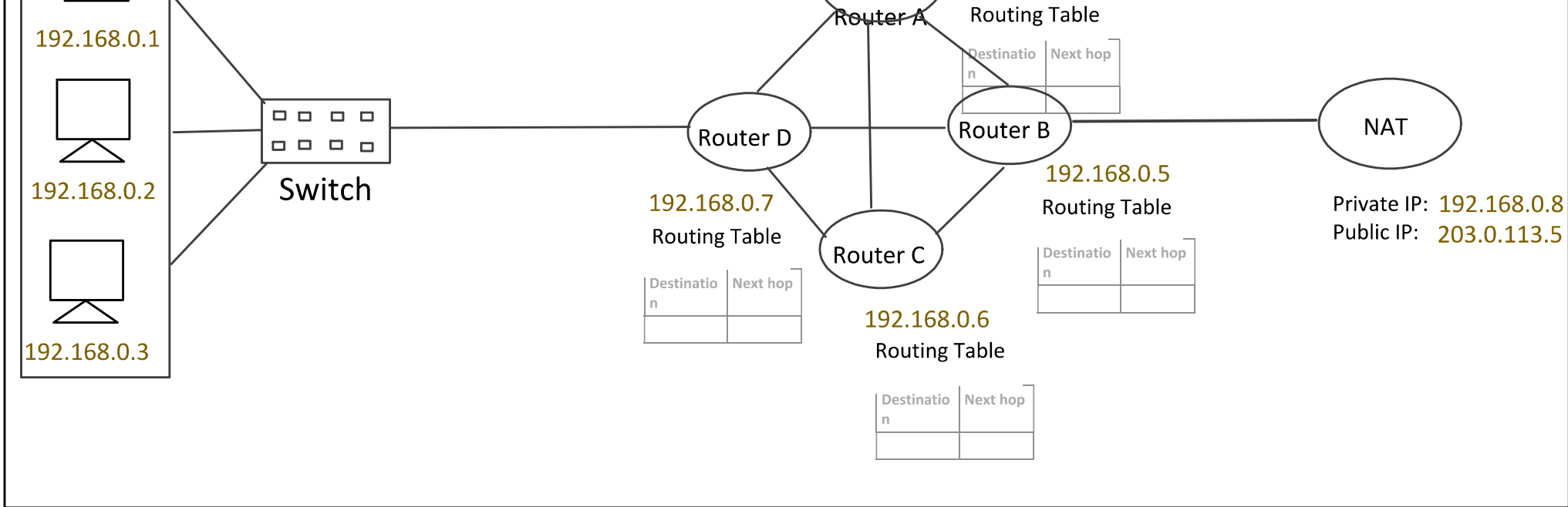
Both are possible scenarios:

1. Routers Mesh in mid

Network:  
CIDR: 192.168.0.0/23

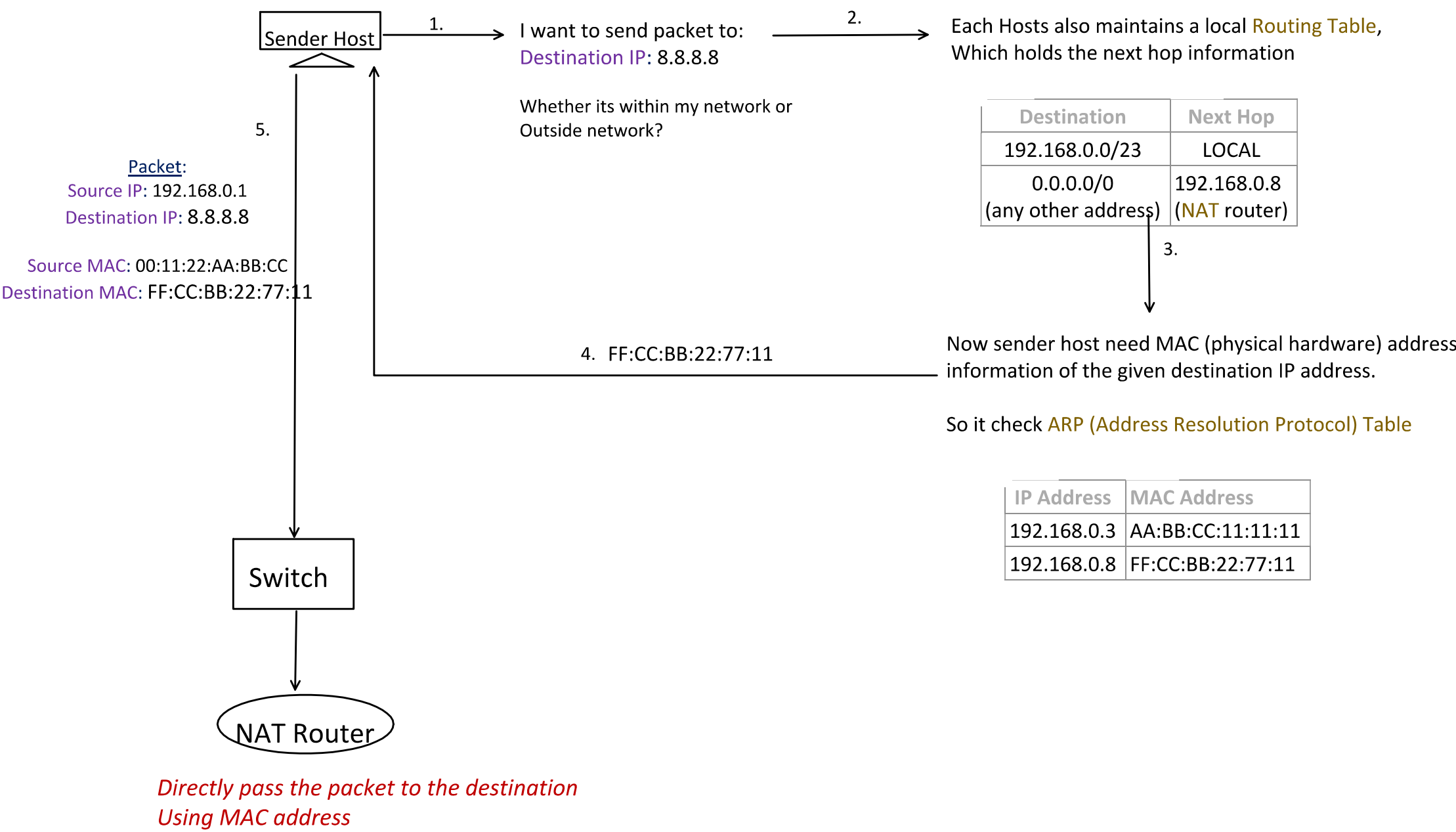
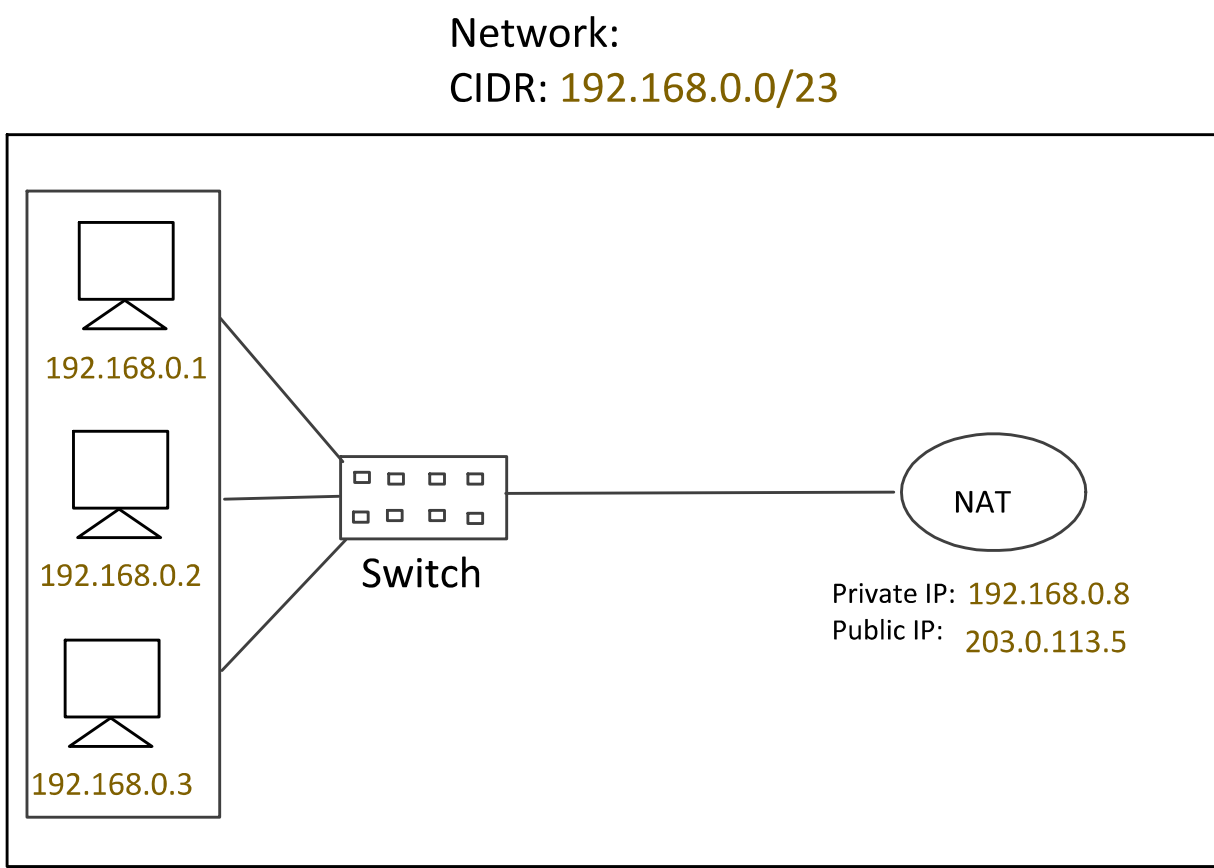




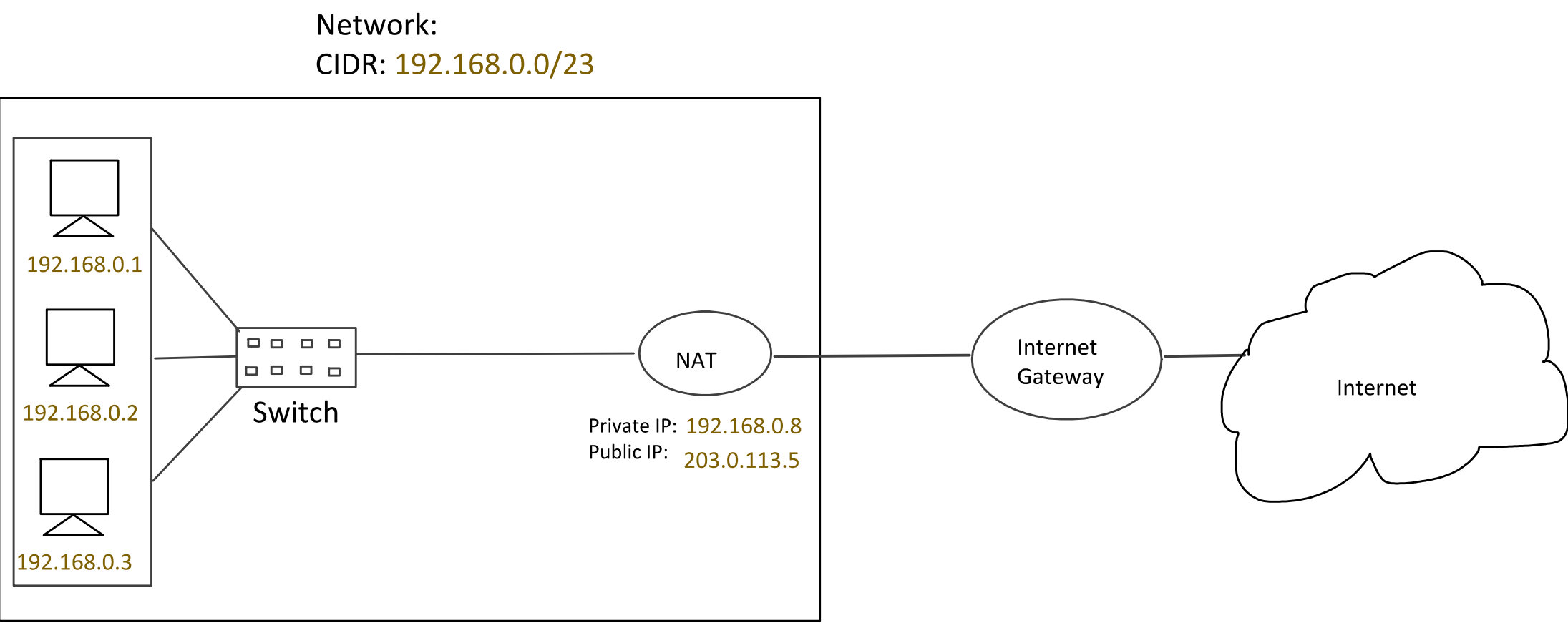


2. No Routers Mesh in Mid:

Small setup, less traffic and no long physical distance, in that scenarios, Switch can directly pass the packet to NAT router.



6. Internet Gateway



OR

